A rigorous Hopf bifurcation in the Becker–Döring bubblelator limit model

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Abstract. Niethammer, Pego, Schlichting and Velázquez derived a limit model for the Becker–Döring equations with injection of monomers and removal of large clusters (the "bubblelator"). Time-periodic solutions of this model solve a scalar integro-differential equation with infinite delay. The authors gave a formal Hopf bifurcation analysis, checked transversality and nonresonance numerically, and wrote that a rigorous Hopf bifurcation theorem had not yet been proven for a model of this type. We prove one. A Lyapunov–Schmidt reduction on spaces of periodic functions, which sidesteps the phase-space problem of infinite delay, shows that three conditions on the explicit function $G_{\beta,\nu}(\lambda) = \int_0^\infty e^{-z^{\beta+1}/(\beta+1)-\lambda z} z^\nu \, dz$ produce a unique smooth branch of periodic solutions, and gives the direction of bifurcation by the formula of the source. A fourth condition makes the bifurcation point the first loss of linear stability. For size-independent removal ($\beta = 0$) all conditions are proved analytically for every $\nu > 0$, and the bifurcation goes in the supercritical direction ($\eta_2 < 0$: the orbits exist where the steady state is unstable) for every $\nu > 0$, by a certified interval computation that includes the limits $\nu \to 0$ and $\nu \to \infty$. For size-dependent removal the conditions and the supercritical direction are certified in Arb ball arithmetic at all 1111 parameter points with $\beta \in \{0, 0.1, \dots, 10\}$ and eleven values of ν: the source's nine values $\alpha = 0.1, \dots, 0.9$ and the physical values $\alpha = 1/3, 2/3$. This is the grid on which the source computed. For $\nu = 0$, $\beta = 3$ the first eleven bifurcation points of the source's Table 1 are certified, with alternating stability of the steady state. Orbital stability of the bifurcating orbits and the full Becker–Döring system are not treated.

1 The question

Niethammer, Pego, Schlichting and Velázquez [1] add a constant source S of monomers and a removal $Rk^r n_k$ of clusters of size $k \geq 2$ to the Becker–Döring equations. In a near-critical regime they derive by formal asymptotics the limit model (their (1.12)–(1.14))

$$\begin{aligned} \partial_\tau u &= 1 - \int_0^\infty x^\alpha f(x,\tau)\, dx, \\ \partial_\tau f + \partial_x(x^\alpha f) &= -\eta\, x^r f \quad (x > 0), \qquad x^\alpha f(x,\tau) \to e^{u(\tau)} \quad (x \to 0), \end{aligned} \tag{1}$$

for a rescaled monomer excess u and a density f of supercritical clusters, with $\alpha \in [0, 1)$, $r \geq 0$ and a rescaled removal rate $\eta > 0$. With $\nu = \alpha/(1-\alpha)$ and $\beta = r/(1-\alpha)$, solutions defined for all times reduce to their equation (3.7),

$$u'(\tau) = 1 - \int_0^\infty e^{u(\tau - z)} \exp\Big(-\frac{\eta}{\beta+1}\, z^{\beta+1}\Big) z^\nu \, dz, \tag{E_η}$$

a scalar equation with infinite delay (see Section 3 for a normalisation remark). In §1.4 (arXiv v2, p. 4) the source states:

> *"To our knowledge, a mathematically rigorous Hopf bifurcation theorem has been proven only for finite delay horizons."*

and at the beginning of §3 (p. 13):

> *"To our knowledge, a rigorous Hopf bifurcation theorem has not yet been proven for a model of this type, though it seems plausible that one might extend existing methods for retarded functional differential equations with finite delays (RFDE) which are based upon rescaling to fix the temporal period and Lyapunov-Schmidt reduction [18, 12]."*

The source then argues "by formal means": it computes the characteristic equation $\vartheta\lambda + G_{\beta,\nu}(\lambda) = 0$, gives the criteria (3.14)–(3.15) for a transversal crossing, derives by a formal expansion the direction coefficients (3.22)–(3.23), proves transversality analytically only for $\beta = 0$ (§4.1), and verifies transversality and the nonresonance condition (3.28) numerically on a grid (§4.2, Appendix C, Tables 2–3; "The transversality condition (3.15) and nonresonance condition (3.28) are verified numerically in the Appendix C").

General Hopf theorems for equations with infinite delay and fading memory do exist [2]; what is missing is a proof for this model, with the hypotheses checked. The present paper supplies it. The proof below is self-contained: it uses only the implicit function theorem on spaces of periodic functions, the route suggested by the source, which goes back to Crandall and Rabinowitz [3].

2 Results

Throughout, $\beta \geq 0$ and $\nu \geq 0$ are fixed. Put

$$G(\lambda) = G_{\beta,\nu}(\lambda) = \int_0^\infty \exp\Big(-\frac{z^{\beta+1}}{\beta+1} - \lambda z\Big) z^\nu \, dz, \qquad H(t) = \operatorname{Re} G(it) \quad (t \in \mathbb{R}),$$

so that $G(0) = (\beta+1)^{(\nu-\beta)/(\beta+1)}\Gamma\big(\frac{\nu+1}{\beta+1}\big)$ and $H'(t) = \operatorname{Im} G_{\beta,\nu+1}(it)$. For $t_0 > 0$ consider the conditions

(H1) $H(t_0) = 0$ and $\operatorname{Im} G(it_0) < 0$; put $\vartheta_0 = -\operatorname{Im} G(it_0)/t_0 > 0$;

(H2) $H'(t_0) \neq 0$;

(H3) $\vartheta_0\, ikt_0 + G(ikt_0) \neq 0$ for every integer $k \geq 2$;

(H4) for every $t > 0$, $t \neq t_0$: if $H(t) = 0$ then $-\operatorname{Im} G(it) < \vartheta_0 t$.

(H4) implies (H3). Under (H1) define the critical removal rate, frequency and the auxiliary numbers

$$\eta_0 = \Big(\frac{\vartheta_0}{G(0)}\Big)^{\beta+1}, \qquad \kappa_0 = \frac{t_0\vartheta_0}{G(0)}, \qquad \mu_0 = \frac{1}{(\beta+1)t_0^{\beta+1}}, \qquad \delta_0 = t_0^{\nu+2}\vartheta_0. \tag{2}$$

Let $\hat g_j(k) = \int_0^\infty e^{-iky}(-y^{\beta+1})^j y^\nu e^{-\mu_0 y^{\beta+1}}\,dy$ and $\ell(k) = ik\delta_0 + \hat g_0(k)$. If (H1)–(H3) hold, define $\mu_2, \delta_2, \eta_2, \kappa_2$ by

$$\mu_2 \operatorname{Re}\hat g_1(1) = -\frac{\delta_0^2 \operatorname{Re}\hat g_0(2)}{4|\ell(2)|^2}, \qquad \delta_2 + \mu_2 \operatorname{Im}\hat g_1(1) = -\frac{\delta_0}{4} + \frac{\delta_0^2(2\delta_0 + \operatorname{Im}\hat g_0(2))}{4|\ell(2)|^2}, \tag{3}$$

$$\eta_2 = (\nu+2)\,\frac{\mu_2}{\mu_0} + (\beta+1)\,\frac{\delta_2}{\delta_0}, \qquad \kappa_2 = \frac{\nu+1}{\beta+1}\,\frac{\mu_2}{\mu_0} + \frac{\delta_2}{\delta_0}. \tag{4}$$

These are the source's formulas (3.22)–(3.23); here they are theorems (Proposition 19).

Theorem 5 (Hopf bifurcation for (E_η))**.** Assume (H1)–(H3) at $t_0 > 0$, and let $u_0(\eta)$ denote the constant solution of (E_η).

(1) $\pm i\kappa_0$ are simple roots of the characteristic function of (E_η) linearised at $u_0(\eta_0)$, and the root $\lambda(\eta)$ continuing $i\kappa_0$ satisfies $\operatorname{sgn} \operatorname{Re} \frac{d\lambda}{d\eta}(\eta_0) = \operatorname{sgn} H'(t_0) \neq 0$. No $ik\kappa_0$, $k \in \mathbb{Z}\backslash\{\pm 1\}$, is a root.

(2) There are $\varepsilon_0 > 0$ and even C^∞ functions $\eta(\varepsilon)$, $\kappa(\varepsilon)$ on $(-\varepsilon_0, \varepsilon_0)$ with $\eta(0) = \eta_0$, $\kappa(0) = \kappa_0$, and for $0 < \varepsilon < \varepsilon_0$ a non-constant C^∞ solution u_ε of $(\mathrm{E}_{\eta(\varepsilon)})$ of minimal period $2\pi/\kappa(\varepsilon)$ with

$$\sup_{\tau\in\mathbb{R}} \left| u_\varepsilon(\tau) - u_0(\eta(\varepsilon)) - \varepsilon \cos(\kappa(\varepsilon)\tau) \right| \leq C\varepsilon^2.$$

(3) There is $\rho > 0$ such that every periodic solution u of (E_η) with period p and $|\eta - \eta_0| + |2\pi/p - \kappa_0| + \|u - u_0(\eta)\|_{C^1(\mathbb{R})} < \rho$ is either constant or a time translate of some u_ε, $0 < \varepsilon < \varepsilon_0$, and then $(\eta, 2\pi/p) = (\eta(\varepsilon), \kappa(\varepsilon))$.

(4) $\eta(\varepsilon) = \eta_0(1 + \eta_2\varepsilon^2 + O(\varepsilon^4))$ and $\kappa(\varepsilon) = \kappa_0(1 + \kappa_2\varepsilon^2 + O(\varepsilon^4))$. In particular, if $\eta_2 < 0$, then for η close to η_0 small periodic solutions near u_0 with frequency near κ_0 exist exactly when $\eta < \eta_0$, and there is one orbit for each such η.

(5) If in addition (H4) holds and $H'(t_0) < 0$, then for every $\eta > \eta_0$ all characteristic roots have negative real part, and for η slightly below η_0 exactly two roots (a complex conjugate pair) have positive real part. Thus η_0 is the largest removal rate at which the steady state loses linear stability.

The same holds for the limit model (1) itself, after the change of normalisation explained in Proposition 9: the critical rate in the variables of (1) is $\eta_0/(1-\alpha)^\beta$, the frequency is κ_0, and f is given along characteristics by (10).

Theorem 6 (size-independent removal, $\beta = 0$)**.** Let $\beta = 0$ and $\nu > 0$. The Hopf points are exactly $t_j = \tan\omega_j$ with $\omega_j = \frac{\pi(1+4j)}{2(\nu+1)}$, $0 \leq j < \nu/4$, at

$$\eta_j = \frac{\cos^{\nu+2}\omega_j}{\sin\omega_j}, \qquad \kappa_j = \cos^{\nu+1}\omega_j.$$

At each of them (H1)–(H3) hold and $H'(t_j) < 0$, so Theorem 5 applies; (H4) holds at t_0. The direction coefficient at the first (largest) point η_0 satisfies $\eta_2 < 0$ for every $\nu > 0$, with $\eta_2 \to -1/36$ as $\nu \to 0$ and $\eta_2 \to \frac{1}{10\pi} - \frac{3}{20}$ as $\nu \to \infty$. Hence, for every $\alpha \in (0,1)$ and $r = 0$, the steady state is linearly stable for $\eta > \eta_0$, unstable for η slightly below η_0, and periodic solutions bifurcate into $\eta < \eta_0$.

Theorem 7 (certified grid, $\beta > 0$)**.** Let $\nu \in \{\frac{1}{9}, \frac{1}{4}, \frac{3}{7}, \frac{1}{2}, \frac{2}{3}, 1, \frac{3}{2}, 2, \frac{7}{3}, 4, 9\}$ (that is $\alpha \in \{0.1, 0.2, 0.3, \frac{1}{3}, 0.4, 0.5, 0.6, \frac{2}{3}, 0.7, 0.8, 0.9\}$) and $\beta \in \{0, \frac{1}{10}, \frac{2}{10}, \ldots, 10\}$. At each of these 1111 points H has a zero t_0 in an explicit interval of width at most $2^{-39}\max(t_0, 1)$ at which (H1), (H2) with $H'(t_0) < 0$, (H4) and $\eta_2 < 0$ hold. By (H4), η_0 is the largest removal rate at which a characteristic root lies on the imaginary

axis. Consequently all conclusions of Theorem 5, including part (5), hold, and the bifurcation is in the direction $\eta < \eta_0$. Enclosures of $\eta_0, \kappa_0, \eta_2, \kappa_2$ are listed in `certificates/grid_results.csv`; a selection is in Table 25.

Theorem 8 ($\nu = 0$, $\beta = 3$)**.** For $\nu = 0$, $\beta = 3$, H has exactly 17 zeros in $(0, 27.77]$, all simple, H' alternates in sign starting with $H'(t_1) < 0$, and $\operatorname{Im} G(it_j) < 0$. The values $\vartheta_j = -\operatorname{Im} G(it_j)/t_j$ decrease strictly for $j = 1, \dots, 11$, and for $\vartheta \geq \vartheta_{11}$ the only purely imaginary roots of $\vartheta\lambda + G(\lambda)$ are $\pm it_j$ at $\vartheta = \vartheta_j$, $j \leq 11$. At each of the eleven points (H1)–(H3) hold, (H4) holds at t_1, and $\eta_2 < 0$. The steady state is linearly stable for $\eta > \eta_{(1)}$, and the number of characteristic roots in the right half-plane alternates between 2 and 0 on the successive intervals between $\eta_{(1)} > \eta_{(2)} > \cdots > \eta_{(11)}$.

3 The limit model as an equation with infinite delay

Proposition 9. Let $u \in C(\mathbb{R})$ be bounded and define, for $x > 0$ and $\tau \in \mathbb{R}$, $a = x^{1-\alpha}/(1-\alpha)$ and

$$x^\alpha f(x,\tau) = e^{u(\tau-a)} \exp\Big(-\eta\,(1-\alpha)^\beta\,\frac{a^{\beta+1}}{\beta+1}\Big). \tag{10}$$

Then f solves the transport equation and the boundary condition of (1) along characteristics, and it is the only such solution defined for all $\tau \in \mathbb{R}$. The first equation of (1) holds if and only if $\tilde{u} = u + \nu \log(1-\alpha)$ solves $(\mathrm{E}_{\tilde{\eta}})$ with $\tilde{\eta} = (1-\alpha)^\beta \eta$.

Proof. The characteristics of $\partial_\tau f + \partial_x(x^\alpha f)$ are $dx/d\tau = x^\alpha$, that is $x^{1-\alpha}/(1-\alpha) = \tau - \sigma$ for the characteristic leaving $x = 0$ at time σ. Along it, $F = x^\alpha f$ satisfies $dF/d\tau = -\eta x^r F$. Since $x^r = ((1-\alpha)(\tau-\sigma))^\beta$, integration from $\sigma = \tau - a$ gives (10). Every backward characteristic reaches $x = 0$ in finite time, so a solution defined for all times is determined by its boundary values. Finally $x = ((1-\alpha)a)^{1/(1-\alpha)}$ gives $dx = ((1-\alpha)a)^\nu da$ and

$$\begin{aligned}\int_0^\infty x^\alpha f\,dx &= (1-\alpha)^\nu \int_0^\infty e^{u(\tau-a)} \exp\Big(-\tilde{\eta}\,\frac{a^{\beta+1}}{\beta+1}\Big) a^\nu\,da \\ &= \int_0^\infty e^{\tilde{u}(\tau-a)} \exp\Big(-\tilde{\eta}\,\frac{a^{\beta+1}}{\beta+1}\Big) a^\nu\,da.\end{aligned}$$

□

Remark 11 (normalisation)**.** The source's change of variables (3.1) produces the transport speed $1-\alpha$ and the boundary value $e^u/(1-\alpha)$, which (3.3)–(3.4) omit. By Proposition 9 the omission only rescales η by $(1-\alpha)^\beta$ and shifts u by a constant. Its equation (3.7), on which its whole Hopf analysis and all its figures and tables rest, is exactly (E_η). We keep its η; for $r = 0$ the two normalisations agree. The direction coefficients η_2, κ_2, being relative, are the same in both. ⌟

Steady states and characteristic function. Put $k_\eta(z) = z^\nu \exp\big(-\eta z^{\beta+1}/(\beta+1)\big)$ and $m(\eta) = \int_0^\infty k_\eta = \eta^{-(\nu+1)/(\beta+1)} G(0)$. The constant solution is $u_0(\eta) = -\log m(\eta)$. Linearising (E_η) at u_0 gives $w'(\tau) = -m(\eta)^{-1} \int_0^\infty k_\eta(z) w(\tau - z)\,dz$, with exponential

solutions $e^{\lambda\tau}$ exactly at the roots of

$$\chi_\eta(\lambda) = \lambda + \frac{1}{m(\eta)}\int_0^\infty k_\eta(z)e^{-\lambda z}dz = \frac{\vartheta\lambda' + G(\lambda')}{G(0)}, \qquad \lambda = \eta^{\frac{1}{\beta+1}}\lambda', \quad \vartheta = \eta^{\frac{1}{\beta+1}}G(0). \tag{12}$$

The function G is entire if $\beta > 0$ and holomorphic in $\operatorname{Re}\lambda > -1$ if $\beta = 0$, and $|G(\lambda)| \le G(0)$, $|G'(\lambda)| \le G_{\beta,\nu+1}(0)$ for $\operatorname{Re}\lambda \ge 0$. A purely imaginary root $\lambda' = it$, $t > 0$, occurs exactly when $H(t) = 0$ and $\vartheta = -\operatorname{Im} G(it)/t > 0$; $\lambda' = 0$ is never a root.

Lemma 13 (source, §3.1). If $\vartheta > G_{\beta,\nu+1}(0)$, then $\vartheta\lambda + G(\lambda) \neq 0$ for $\operatorname{Re}\lambda \ge 0$.

Proof. If $\vartheta\lambda + G(\lambda) = 0$ with $\operatorname{Re}\lambda \ge 0$, then $\lambda \neq 0$ and $\vartheta = \operatorname{Re}\frac{G(0)-G(\lambda)}{\lambda} - G(0)\operatorname{Re}\frac{1}{\lambda} \le \left|\int_0^1 G'(s\lambda)ds\right| \le G_{\beta,\nu+1}(0)$. □

4 A Hopf theorem for (E_η)

4.1 Periodic reformulation

Let $\kappa > 0$ and let u be a $2\pi/\kappa$-periodic C^1 solution of (E_η). Put

$$U(s) = u(s/\kappa) - u_0(\eta), \qquad v = e^U - 1,$$
$$\mu = \frac{\eta}{(\beta+1)\kappa^{\beta+1}}, \qquad \delta = \kappa^{\nu+2}m(\eta), \qquad g_\mu(y) = y^\nu e^{-\mu y^{\beta+1}}.$$

Since $e^{u_0}m = 1$, (E_η) reads $\kappa U'(s) = -e^{u_0}\int_0^\infty k_\eta(z)(e^{U(s-\kappa z)} - 1)dz$; the substitution $y = \kappa z$ and $k_\eta(y/\kappa) = \kappa^{-\nu}g_\mu(y)$ turn it into $\delta U' = -g_\mu * v$, i.e.

$$F(v,\delta,\mu) := \delta v' + (1+v)(g_\mu * v) = 0, \qquad (g_\mu * v)(s) = \int_0^\infty g_\mu(y)v(s-y)\,dy, \tag{14}$$

for the 2π-periodic function $v > -1$. Conversely every 2π-periodic C^1 solution $v > -1$ of (14) gives a $2\pi/\kappa$-periodic solution $u(\tau) = u_0(\eta) + \log(1 + v(\kappa\tau))$ of (E_η), where

$$\kappa = \frac{\delta\,((\beta+1)\mu)^{\frac{\nu+1}{\beta+1}}}{G(0)}, \qquad \eta = (\beta+1)\mu\kappa^{\beta+1} = \frac{(\beta+1)^{\nu+2}\mu^{\nu+2}\delta^{\beta+1}}{G(0)^{\beta+1}}. \tag{15}$$

The map $(\kappa,\eta) \mapsto (\delta,\mu)$ is a real-analytic diffeomorphism of $(0,\infty)^2$ with inverse (15). Fourier coefficients are $\hat v(k) = \frac{1}{2\pi}\int_0^{2\pi} v(s)e^{-iks}ds$, and by Fubini $(g_\mu * v)\hat{}\,(k) = \hat g_\mu(k)\hat v(k)$ with $\hat g_\mu(k) = \int_0^\infty g_\mu(y)e^{-iky}dy$.

Lemma 16. At (δ_0,μ_0) from (2) one has $\hat g_{\mu_0}(k) = t_0^{\nu+1}G(ikt_0)$, hence $\ell(k) = t_0^{\nu+1}(\vartheta_0\, ikt_0 + G(ikt_0))$, and

$$\operatorname{Re}\partial_\mu\hat g_\mu(1)\big|_{\mu_0} = -t_0^{\nu+\beta+3}H'(t_0).$$

Thus (H1), (H2), (H3) are equivalent to: (A1) $\ell(1) = 0$; (A2) $\operatorname{Re}\hat g_1(1) \neq 0$; (A3) $\ell(k) \neq 0$ for $k \ge 2$. Moreover (δ_0,μ_0) corresponds to (κ_0,η_0) under (15).

Proof. The first identity is the substitution $y = t_0 z$ together with $\mu_0 t_0^{\beta+1} = 1/(\beta+1)$, and the formula for $\ell(k)$ follows from $\delta_0 = t_0^{\nu+2}\vartheta_0$. For the last one, integrate $\frac{d}{dy}[y^{\nu+1}e^{-\mu y^{\beta+1}}]e^{-iy}$ by parts (the boundary terms vanish):

$$(\nu+1)\hat g_\mu(1) + (\beta+1)\mu\,\partial_\mu\hat g_\mu(1) = i\int_0^\infty y^{\nu+1}e^{-\mu y^{\beta+1}}e^{-iy}dy.$$

At μ_0, $\hat g_{\mu_0}(1) = -i\delta_0$ is imaginary under (H1); the right side equals $i\,t_0^{\nu+2} G_{\beta,\nu+1}(it_0)$, whose real part is $-t_0^{\nu+2} H'(t_0)$. Taking real parts and using $(\beta+1)\mu_0 = t_0^{-\beta-1}$ gives the claim. The correspondence of the critical points is a direct substitution into (15). □

4.2 Functional setting

Let $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, $X = H^1(\mathbb{T};\mathbb{R})$, $Y = L^2(\mathbb{T};\mathbb{R})$. For $\mu > 0$, $g_\mu \in L^1(0,\infty)$ and $C_\mu v = g_\mu * v$ is bounded on Y and on X with norm at most $\|g_\mu\|_{L^1}$ (Minkowski's inequality); it commutes with translations and with d/ds.

Lemma 17. $F : X \times (0,\infty)^2 \to Y$ in (14) is C^∞, $F(0,\delta,\mu) = 0$, and $D_v F(0,\delta,\mu) = L(\delta,\mu) = \delta\, d/ds + C_\mu$ is the Fourier multiplier $\ell(k;\delta,\mu) = ik\delta + \hat g_\mu(k)$. F is equivariant: $F(v(\cdot+\sigma),\delta,\mu) = F(v,\delta,\mu)(\cdot+\sigma)$.

Proof. For $0 < \mu_1 \le \mu$, $|\partial_\mu^j g_\mu(y)| = y^{j(\beta+1)+\nu} e^{-\mu y^{\beta+1}} \le y^{j(\beta+1)+\nu} e^{-\mu_1 y^{\beta+1}} \in L^1$, so $\mu \mapsto g_\mu \in L^1(0,\infty)$ is C^∞ by dominated convergence and the mean value theorem, hence so is $\mu \mapsto C_\mu \in \mathcal{B}(Y)$. The map $(v,\delta) \mapsto \delta v'$ is bilinear and bounded $X \times \mathbb{R} \to Y$, and $(v,w) \mapsto vw$ is bilinear and bounded $X \times Y \to Y$ because $\|v\|_\infty \le c\|v\|_{H^1}$. The rest is immediate. □

4.3 Lyapunov–Schmidt reduction

Let P be the projection onto the modes $k = \pm 1$ (on X and on Y) and $Q = I - P$.

Proposition 18. Assume (A1)–(A3) at (δ_0,μ_0). There are $r_0 > 0$, a neighbourhood $\mathcal{U}$ of $(0,\delta_0,\mu_0)$ in $X \times \mathbb{R}^2$, and C^∞ maps $r \mapsto (W(r),\delta^*(r),\mu^*(r))$ on $(-r_0,r_0)$, with $W(0) = 0$, $PW(r) = 0$, δ^*, μ^* even, $\delta^*(0) = \delta_0$, $\mu^*(0) = \mu_0$, such that

(1) $v_r = r(e^{is} + e^{-is}) + W(r) = 2r\cos s + W(r)$ solves $F(v_r,\delta^*(r),\mu^*(r)) = 0$, and $\|W(r)\|_X = O(r^2)$;

(2) every solution $(v,\delta,\mu) \in \mathcal{U}$ of $F = 0$ has either $v = 0$, or $v = v_r(\cdot+\sigma)$ for some $r \in (0,r_0)$, $\sigma \in \mathbb{R}$, and $(\delta,\mu) = (\delta^*(r),\mu^*(r))$.

Proof. *Step 1: the range equation.* Since $|\hat g_{\mu_0}(k)| \le \|g_{\mu_0}\|_{L^1}$, $|\ell(k)| \ge |k|\delta_0/2$ for $|k| \ge 2\|g_{\mu_0}\|_{L^1}/\delta_0$, and (A3) together with $\ell(0) = \hat g_{\mu_0}(0) > 0$ and $\ell(-k) = \overline{\ell(k)}$ shows that $\ell(k) \ne 0$ for all $k \ne \pm 1$. Hence $\sup_{k\ne\pm1}(1+|k|)/|\ell(k)| < \infty$ and $QL(\delta_0,\mu_0) : QX \to QY$ is an isomorphism, with inverse the multiplier $1/\ell(k)$. Write $v = ae^{is} + \bar a e^{-is} + w$, $a \in \mathbb{C}$, $w \in QX$. The implicit function theorem applied to $QF(ae^{is} + \bar a e^{-is} + w,\delta,\mu) = 0$ at $(0,0,\delta_0,\mu_0)$ gives a C^∞ solution $w = W(a,\delta,\mu)$ on a set $\{|a| < a_0, |\delta-\delta_0| < \rho, |\mu-\mu_0| < \rho\}$, unique in $\{\|w\|_X < \rho_w\}$. Since $F(0,\delta,\mu) = 0$, $W(0,\delta,\mu) = 0$; differentiating at $a = 0$ and using $QLe^{\pm is} = 0$ gives $D_a W(0,\delta,\mu) = 0$, so $W = O(|a|^2)$. As translations are isometries of X that multiply $\hat v(1)$ by $e^{i\sigma}$ and commute with P, uniqueness gives $W(e^{i\sigma}a,\delta,\mu) = W(a,\delta,\mu)(\cdot+\sigma)$.

Step 2: the bifurcation equation. Let $\Phi(a,\delta,\mu)$ be the coefficient of e^{is} in $F(ae^{is} + \bar a e^{-is} + W(a,\delta,\mu),\delta,\mu)$; the coefficient of e^{-is} is its conjugate. Equivariance gives $\Phi(e^{i\sigma}a,\cdot) = e^{i\sigma}\Phi(a,\cdot)$. For real r, $\Phi(0,\delta,\mu) = 0$, so $\Phi(r,\delta,\mu) = r\Psi(r,\delta,\mu)$ with $\Psi = \int_0^1 \partial_r \Phi(\theta r,\delta,\mu)\,d\theta$ of class C^∞, and $\Psi(0,\delta,\mu) = \ell(1;\delta,\mu) = i\delta + \hat g_\mu(1)$ because $D_a W(0) = 0$. By (A1) $\Psi(0,\delta_0,\mu_0) = 0$, and the real Jacobian of $(\delta,\mu) \mapsto i\delta + \hat g_\mu(1)$ at (δ_0,μ_0) has determinant $-\operatorname{Re}\hat g_1(1) \ne 0$ by (A2). The implicit function theorem

gives C^∞ functions $(\delta^*,\mu^*)(r)$, $|r|<r_0$, with $\Psi(r,\delta,\mu)=0$ near $(0,\delta_0,\mu_0)$ if and only if $(\delta,\mu)=(\delta^*(r),\mu^*(r))$. Since $\Phi(-r)=\Phi(e^{i\pi}r)=-\Phi(r)$, Ψ is even in r, and so are δ^*,μ^*. Put $W(r)=W(r,\delta^*(r),\mu^*(r))$.

Step 3: all small solutions. Take $\mathcal{U}$ so small that $(\hat v(1),Qv,\delta,\mu)$ lies in the domains above. If $F(v,\delta,\mu)=0$, then $Qv=W(\hat v(1),\delta,\mu)$. If $\hat v(1)=0$ then $v=W(0,\delta,\mu)=0$. Otherwise $\hat v(1)=re^{i\sigma}$ with $r>0$; the translate $v(\cdot-\sigma)$ has first coefficient r, so $\Psi(r,\delta,\mu)=0$, $(\delta,\mu)=(\delta^*(r),\mu^*(r))$ and $v(\cdot-\sigma)=v_r$. □

Regularity. A solution $v\in H^1\subset C(\mathbb{T})$ of (14) has $v'=-(1+v)(g_\mu*v)/\delta\in C(\mathbb{T})$; inductively $v\in C^m$ implies $g_\mu*v\in C^m$ and $v\in C^{m+1}$, so $v\in C^\infty$.

4.4 Direction of bifurcation

Proposition 19. Under (A1)–(A3), with $\varepsilon=2r$, $\delta^*=\delta_0+\delta_2\varepsilon^2+O(\varepsilon^4)$ and $\mu^*=\mu_0+\mu_2\varepsilon^2+O(\varepsilon^4)$, where μ_2,δ_2 are given by (3). The corresponding (κ,η) satisfy $\eta=\eta_0(1+\eta_2\varepsilon^2+O(\varepsilon^4))$, $\kappa=\kappa_0(1+\kappa_2\varepsilon^2+O(\varepsilon^4))$ with (4).

Proof. The map $\varepsilon\mapsto(v_{\varepsilon/2},\delta^*,\mu^*)\in X\times\mathbb{R}^2$ is C^∞, and δ^*,μ^* are even, so $v=\varepsilon v_1+\varepsilon^2v_2+\varepsilon^3v_3+O(\varepsilon^4)$ in X with $v_1=\cos s$ and $Pv_j=0$ for $j\ge2$, and $C_{\mu^*}=C_0+\mu_2\varepsilon^2C_1+O(\varepsilon^4)$ in $\mathcal{B}(Y)$, where C_j is convolution with $g_j=\partial_\mu^j g_\mu|_{\mu_0}$. Expanding the identity $F(v,\delta^*,\mu^*)=0$ (a C^∞ function of ε) and writing $L=L(\delta_0,\mu_0)$:

$$\varepsilon:\ Lv_1=0;\qquad \varepsilon^2:\ Lv_2+v_1C_0v_1=0;$$
$$\varepsilon^3:\ Lv_3+\delta_2v_1'+\mu_2C_1v_1+v_2C_0v_1+v_1C_0v_2=0.$$

By (A1), $C_0\cos s=\operatorname{Re}(\hat g_0(1)e^{is})=\delta_0\sin s$, so $v_1C_0v_1=\frac{\delta_0}{2}\sin2s$ and $v_2=\hat v_2(2)e^{2is}+\text{c.c.}$ with $\hat v_2(2)=\frac{i\delta_0}{4\ell(2)}$ ($\ell(\pm2)\neq0$ by (A3)). The coefficient of e^{is} in the ε^3 equation vanishes for Lv_3 (since $\ell(1)=0$) and gives

$$\tfrac12\left(i\delta_2+\mu_2\hat g_1(1)\right)+\tfrac12\,\hat v_2(2)\,\overline{\hat g_0(1)}+\tfrac12\,\hat g_0(2)\hat v_2(2)=0.$$

Since $\overline{\hat g_0(1)}=i\delta_0$ and $i\delta_0+\hat g_0(2)=\ell(2)-i\delta_0$, this is

$$i\delta_2+\mu_2\hat g_1(1)+\frac{i\delta_0}{4}+\frac{\delta_0^2}{4\ell(2)}=0.$$

Real and imaginary parts are (3). Finally (15) gives $\eta/\eta_0=(\mu/\mu_0)^{\nu+2}(\delta/\delta_0)^{\beta+1}$ and $\kappa/\kappa_0=(\delta/\delta_0)(\mu/\mu_0)^{(\nu+1)/(\beta+1)}$, whence (4). □

4.5 Characteristic roots

Proposition 20. Assume (H1), (H2). Then it_0 is a simple root of $\vartheta_0\lambda+G(\lambda)$ and its continuation $\lambda'(\vartheta)$ satisfies $\operatorname{Re}\frac{d\lambda'}{d\vartheta}(\vartheta_0)=t_0H'(t_0)/|\vartheta_0+G'(it_0)|^2$. If also (H4) holds, then for $\vartheta>\vartheta_0$ the equation $\vartheta\lambda+G(\lambda)=0$ has no root with $\operatorname{Re}\lambda\ge0$, and if moreover $H'(t_0)<0$ it has exactly two roots with $\operatorname{Re}\lambda>0$ for ϑ slightly below ϑ_0.

Proof. Write $\Lambda(\lambda,\vartheta)=\vartheta\lambda+G(\lambda)$. Since $\frac{d}{dt}G(it)=iG'(it)$, $H'(t)=-\operatorname{Im}G'(it)$, so $\operatorname{Im}\partial_\lambda\Lambda(it_0,\vartheta_0)=-H'(t_0)\neq0$: the root is simple. Then $\frac{d\lambda'}{d\vartheta}=-it_0/\partial_\lambda\Lambda$, whose real part is $-t_0\operatorname{Im}\partial_\lambda\Lambda/|\partial_\lambda\Lambda|^2=t_0H'(t_0)/|\partial_\lambda\Lambda|^2$.

Let $N(\vartheta)$ be the number of roots in $\operatorname{Re}\lambda > 0$. All roots with $\operatorname{Re}\lambda \geq 0$ satisfy $|\lambda| \leq G(0)/\vartheta$. By (H4), for $\vartheta > \vartheta_0$ there is no root on $i\mathbb{R}$: a root it ($t > 0$; $t < 0$ is the conjugate case and $\lambda = 0$ is impossible) needs $H(t) = 0$ and $-\operatorname{Im} G(it) = \vartheta t > \vartheta_0 t$, excluded by (H4) for $t \neq t_0$, and at $t = t_0$ the value of ϑ is ϑ_0. On a compact interval $[a, b] \subset (\vartheta_0, \infty)$, choose $M > G(0)/a$; there are no roots on the boundary of the half disc $\{\operatorname{Re}\lambda > 0, |\lambda| < M\}$, so by the argument principle N is continuous, hence constant, on (ϑ_0, ∞). By Lemma 13, $N = 0$ for large ϑ, so $N = 0$ on (ϑ_0, ∞). At ϑ_0 there is no root with $\operatorname{Re}\lambda > 0$ (by Hurwitz's theorem it would persist for ϑ slightly above ϑ_0), and by (H4) the only imaginary roots are $\pm it_0$, both simple. By Hurwitz's theorem, for ϑ close to ϑ_0 the roots in the closed half disc lie in small discs around $\pm it_0$, one in each, namely $\lambda'(\vartheta)$ and $\overline{\lambda'(\vartheta)}$. If $H'(t_0) < 0$, $\operatorname{Re}\lambda'(\vartheta) > 0$ for $\vartheta < \vartheta_0$ close to ϑ_0, so $N = 2$. □

Proof (Proof of Theorem 5). *Part (1).* By (12), the roots of χ_η are $\lambda = (\vartheta/G(0))\lambda'$, where $\vartheta = \eta^{1/(\beta+1)}G(0)$ is increasing in η; at η_0 the root $\lambda' = it_0$ gives $\lambda = i\kappa_0$. Since $d\lambda/d\vartheta = \lambda'/G(0) + (\vartheta/G(0))\, d\lambda'/d\vartheta$ and λ' is imaginary, Proposition 20 gives the sign. A root $ik\kappa_0$ with $k \geq 2$ would mean $\ell(k) = 0$ (Lemma 16), excluded by (H3); $k = 0$ is excluded since $\chi_\eta(0) = 1$, and negative k by conjugation.

Parts (2)–(3). Lemma 16 turns (H1)–(H3) into (A1)–(A3). Apply Proposition 18 and transport the result by (15): $\eta(\varepsilon), \kappa(\varepsilon)$ are the images of $(\delta^*, \mu^*)(\varepsilon/2)$, and $u_\varepsilon(\tau) = u_0(\eta(\varepsilon)) + \log\bigl(1 + v_{\varepsilon/2}(\kappa(\varepsilon)\tau)\bigr)$. Since $v_{\varepsilon/2} = \varepsilon\cos s + O(\varepsilon^2)$ in $X \subset C(\mathbb{T})$, the estimate follows. The minimal period of $v_{\varepsilon/2}$ is 2π, because its first Fourier coefficient is $\varepsilon/2 \neq 0$. If u is a periodic solution as in part (3), then $v(s) = e^{u(s/\kappa)-u_0(\eta)} - 1$, $\kappa = 2\pi/p$, has small $C^1(\mathbb{T})$ norm, hence small X norm, and (δ, μ) is close to (δ_0, μ_0), so part (2) of Proposition 18 applies.

Part (4) is Proposition 19. If $\eta_2 < 0$, then $\varepsilon \mapsto \eta(\varepsilon)$ is strictly decreasing on $(0, \varepsilon_1)$ for some $\varepsilon_1 > 0$, which with part (3) gives the last statement.

Part (5) is Proposition 20 transported by (12); (H4) implies (H3) because a root ikt_0 at ϑ_0 would be a zero $t = kt_0 \neq t_0$ of H with $-\operatorname{Im} G(it) = \vartheta_0 t$. □

Remark 21. Theorem 5 concerns existence, uniqueness and direction. Orbital stability of u_ε needs a phase space for (E_η) and a centre manifold or Floquet theory with infinite delay (fading-memory spaces [2] [4], or the sun-star framework [5]); we do not prove it. The source expects stability when $\eta_2 < 0$ at the largest bifurcation point, by analogy with ordinary and finite-delay equations [6]. ⌟

5 Size-independent removal: proof of Theorem 6

For $\beta = 0$, $G(\lambda) = \Gamma(N)(1+\lambda)^{-N}$ with $N = \nu + 1$ (source, §4.1). Then $H(t) = \Gamma(N)(1+t^2)^{-N/2}\cos(N\arctan t)$ and $-\operatorname{Im} G(it) = \Gamma(N)(1+t^2)^{-N/2}\sin(N\arctan t)$.

Hopf points. $H(t) = 0$ and $\operatorname{Im} G(it) < 0$ mean $N\arctan t \equiv \pi/2 \pmod{2\pi}$, i.e. $\arctan t = \omega_j$ with $\omega_j < \pi/2$, i.e. $0 \leq 4j < \nu$. With $t_j = \tan\omega_j$, $(1+t_j^2)^{-1/2} = \cos\omega_j$, so $\vartheta_j = \Gamma(N)\cos^{N+1}\omega_j/\sin\omega_j$ and, since $G(0) = \Gamma(N)$, (2) gives the stated η_j, κ_j.

(H2). $H'(t_j) = -N\Gamma(N)(1+t_j^2)^{-N/2-1}\sin(N\arctan t_j) = -N\Gamma(N)\cos^{N+2}\omega_j < 0$.

(H3), (H4). At any imaginary root it for the value ϑ, $|G(it)| = \vartheta t$, i.e. $\Gamma(N)(1+t^2)^{-N/2} = \vartheta t$; the left side decreases and the right side increases in t, so for a given ϑ there is at most one $t > 0$. This gives (H3) at every t_j. For (H4) at t_0: if $H(t) = 0$ and $t \neq t_0$,

either $\operatorname{Im} G(it) > 0$ and $-\operatorname{Im} G(it) < 0 < \vartheta_0 t$, or $t = t_j$ with $j \geq 1$, so $t > t_0$ and $-\operatorname{Im} G(it) = \Gamma(N)(1+t^2)^{-N/2} < \Gamma(N)(1+t_0^2)^{-N/2} = \vartheta_0 t_0 < \vartheta_0 t$.

Direction at η_0. Write $\omega = \omega_0 = \pi/(2N)$, $s = \sin\omega$, $c = \cos\omega$. Here $\mu_0 = \cot\omega$, $\hat{g}_\mu(k) = \Gamma(N)(\mu + ik)^{-N}$, and $\cot\omega + i = e^{i\omega}/s$, $\cot\omega + 2i = (c+2is)/s$, $e^{-iN\omega} = -i$. Hence $\delta_0 = \Gamma(N)s^N$, $\hat{g}_0(2) = \Gamma(N)s^N A$ with $A = (c+2is)^{-N}$ (principal power), and $\hat{g}_1(1) = -N\Gamma(N)(\mu_0 + i)^{-N-1} = N\Gamma(N)s^{N+1}(s+ic)$, so $\operatorname{Re}\hat{g}_1(1) > 0$ for every $\nu > 0$. Inserting these into (3)–(4) and using $(N+1)\tan\omega - Nsc = \tan\omega\,(1+Ns^2)$:

Proposition 22. For $\beta = 0$, with $D = |2i + A|^2$,

$$\eta_2 = -\frac{\operatorname{Re} A}{c}\,\frac{1+Ns^2}{4NsD} - \frac{1}{4} + \frac{2+\operatorname{Im} A}{4D}, \qquad \kappa_2 = -\frac{\operatorname{Re} A\,\tan\omega}{4D} - \frac{1}{4} + \frac{2+\operatorname{Im} A}{4D}.$$

For $\nu = 1$: $\eta_2 = -1/16$, $\kappa_2 = -1/12$.

Lemma 23. $\eta_2 < 0$ for every $\nu > 0$; $\eta_2 \to -1/36$ as $\nu \to 0$ and $\eta_2 \to \frac{1}{10\pi} - \frac{3}{20}$ as $\nu \to \infty$.

Proof (Proof (computer-assisted, script `01_beta0_direction.py`)). Parametrise by $\omega \in (0, \pi/2)$, $N = \pi/(2\omega)$. On $[0.05, \pi/2 - 0.05]$ the formula is evaluated in ball arithmetic on 56 subintervals (adaptive bisection), each giving an enclosure in $(-\infty, 0)$. Near the ends the formula has removable singularities, which we remove explicitly.

$\omega \in [0, 0.05]$ ($\nu \geq 30.4$). $Ns = \frac{\pi}{2}\operatorname{sinc}\omega$, $Ns^2 = Ns \cdot s$, and $A = \exp(-N\log(c+2is))$ with $N\log(c+2is) = \frac{\pi}{4}\,3s\operatorname{sinc}\omega\operatorname{Lg}(3s^2) + i\pi\operatorname{tanc}\omega\operatorname{atanc}(2\tan\omega)$, where $\operatorname{sinc} x = \frac{\sin x}{x}$, $\operatorname{tanc} x = \frac{\tan x}{x}$, $\operatorname{Lg}(y) = \frac{\log(1+y)}{y}$, $\operatorname{atanc} x = \frac{\arctan x}{x}$, all extended by 1 at 0. These are monotone (sinc decreasing on $[0,\pi]$, tanc increasing on $[0,\pi/2)$, Lg and atanc decreasing on $[0,\infty)$), so their ranges on an interval are given by the endpoint values.

$\epsilon = \pi/2 - \omega \in [0, 0.05]$ ($\nu \leq 0.033$). Then $N = (1 - 2\epsilon/\pi)^{-1}$, $c = \sin\epsilon$, $s = \cos\epsilon$, $c + 2is = 2i(\cos\epsilon - \frac{i}{2}\sin\epsilon)$, and $A = -i\,2^{-N}e^W$ with $\operatorname{Re} W = -\frac{N}{2}\log\!\left(1 - \frac{3}{4}\sin^2\epsilon\right)$ and $\operatorname{Im} W = q\epsilon$, $q = -N(1 - \frac{1}{2}\operatorname{tanc}\epsilon\operatorname{atanc}(\frac{1}{2}\tan\epsilon))$ (using $(N-1)\pi/2 = N\epsilon$). Thus $\operatorname{Re} A/c = 2^{-N}e^{\operatorname{Re} W}\operatorname{sinc}(q\epsilon)\,q/\operatorname{sinc}\epsilon$ is regular at $\epsilon = 0$.

In both end regions a single ball evaluation over the whole region already lies in $(-\infty, 0)$; the values at the end points give the two limits (at $\epsilon = 0$: $N = 1$, $q = -\frac{1}{2}$, $A = -\frac{i}{2}$, $\operatorname{Re} A/c = -\frac{1}{4}$, $D = \frac{9}{4}$, $\eta_2 = \frac{1}{18} - \frac{1}{12}$). The closed form was also checked against the general integral-based routine of Section 6 at $\nu = \frac{1}{9}, \frac{1}{2}, 1, 9$. □

This proves Theorem 6, with Proposition 20 for the stability statements.

6 Certified verification for $\beta > 0$

6.1 Enclosing G

Lemma 24. Let λ lie in the closed first quadrant and $0 < \varphi$, $(\beta+1)\varphi < \pi/2$. Then

$$G_{\beta,\nu}(\lambda) = e^{-i(\nu+1)\varphi}\int_0^\infty \exp\!\Big(-\frac{r^{\beta+1}e^{-i(\beta+1)\varphi}}{\beta+1} - \lambda r e^{-i\varphi}\Big) r^\nu\,dr,$$

and the integrand is bounded by $r^\nu e^{-cr^{\beta+1}}$, $c = \cos((\beta+1)\varphi)/(\beta+1) > 0$. Con-

sequently, for $0 < \epsilon < 1 < R$,

$$\Big|\int_0^{\epsilon}\Big| \le \frac{\epsilon^{\nu+1}}{\nu+1}, \qquad \Big|\int_R^{\infty}\Big| \le \frac{c^{-s}\,\Gamma(s, cR^{\beta+1})}{\beta+1}, \quad s = \frac{\nu+1}{\beta+1}.$$

Proof. The integrand $\exp\!\big(-z^{\beta+1}/(\beta+1) - \lambda z\big) z^{\nu}$ (principal powers) is holomorphic in the slit plane. For $z = re^{-i\theta}$, $0 \le \theta \le \varphi$, $\operatorname{Re}(\lambda z) = r(\operatorname{Re}\lambda\cos\theta + \operatorname{Im}\lambda\sin\theta) \ge 0$ and $\operatorname{Re} z^{\beta+1} = r^{\beta+1}\cos((\beta+1)\theta) \ge (\beta+1)cr^{\beta+1}$, so on the arc $|z| = R$ the integrand is at most $R^{\nu}e^{-cR^{\beta+1}}$ and on the small arc at most ϵ^{ν}. Cauchy's theorem in the sector and $R \to \infty$, $\epsilon \to 0$ give the formula, and the bounds follow by integrating the majorant. □

We take for φ the double-precision number nearest to $0.75/(\beta+1)$ (the program checks $c > 0$ in ball arithmetic). The integral is split into the dyadic pieces $[2^{-k-1}, 2^{-k}]$, $1 \le k < K$, where K is chosen so that $2^{-K(\nu+1)}/(\nu+1)$ is below the target tolerance, and the pieces $[\frac{1}{2}, 1], [1, 2], [2, 4], \ldots, [\cdot, R]$. Each piece is computed by the rigorous Gauss–Legendre integrator of Arb [7] [8], called through python-flint, with the branch-cut-aware ("analytic") versions of $r^{\beta+1}$ and r^{ν} that the integrator requires. Parameters may be balls: if t ranges over a ball, the output encloses $G(it)$ for every t in it. Precision is 128 bits, the tolerance 2^{-70} (2^{-90} in Section 7). For $\beta = 0$ the routine was checked against the closed form, and against the source's values (Section 6.3).

6.2 The certificate at one parameter point

For β, ν (exact rationals, enclosed in balls) and an interval $T = [t_{\mathrm{lo}}, t_{\mathrm{hi}}]$ found by a floating-point search (not part of the proof), script `03_certify_grid.py` checks:

(C1) $H(t_{\mathrm{lo}}) > 0 > H(t_{\mathrm{hi}})$;

(C2) $H' < 0$ on $T' = [t_{\mathrm{lo}} - d, t_{\mathrm{hi}} + d]$ (16 sub-balls; d starts at $0.05\,t_{\mathrm{mid}}$ and is halved on failure);

(C3) $\operatorname{Im} G(it) < 0$ for $t \in T$; let $\underline{\vartheta}$ be a lower bound of $-\operatorname{Im} G(iT)/T$;

(C4) with $T_{\max} = G(0)/\underline{\vartheta}$: $G(0) - 2^{-20}G_{\beta,\nu+1}(0) > 0$, and $(2^{-20}, T_{\max}]\backslash T'$ is covered by 128 intervals, bisected when needed, on each of which either $\operatorname{Re} G(it)$ has constant sign or $\sup(-\operatorname{Im} G(it)) < \underline{\vartheta} \cdot \inf t$;

(C5) enclosures of $\operatorname{Re}\hat{g}_1(1)$, $|\ell(2)|$, $\eta_0, \kappa_0, \eta_2, \kappa_2$ from (2)–(4) with $t \in T$, via Lemma 16; $\operatorname{Re}\hat{g}_1(1) > 0$, $\ell(2) \ne 0$, and the sign of η_2.

Why these suffice. By (C1) and (C2), H has exactly one zero t_0 in T', it lies in T, and $H'(t_0) < 0$: this is (H2) and the zero part of (H1); (C3) gives the rest of (H1). For (H4), let $t \ne t_0$ with $H(t) = 0$. If $t \le 2^{-20}$: $|G(it) - G(0)| \le tG_{\beta,\nu+1}(0)$, so $H(t) > 0$, impossible. If $t \in T'$: impossible. If t lies in a scanned interval J: $H \ne 0$ on J unless $-\operatorname{Im} G(it) < \underline{\vartheta} \inf J \le \vartheta_0 t$. If $t > T_{\max}$: $-\operatorname{Im} G(it) \le |G(it)| \le G(0) < \underline{\vartheta} t \le \vartheta_0 t$. Every enclosure in (C5) contains the value at the true t_0 because $t_0 \in T$.

6.3 Results

All 1111 points of Theorem 7 pass (C1)–(C5) with $\eta_2 < 0$; this proves Theorem 7. The run takes about three minutes on eight cores. Table 25 lists a selection. The enclosures agree with the floating-point values of the source (the two data files of its arXiv source

give $\eta_0, \eta_2, \kappa_0, \kappa_2, \operatorname{Re}\hat{g}_1(1), |\ell(2)|$ for its nine values of ν and $\beta = 0, 0.1, \dots, 10$): over these 909 points the largest relative deviation is $7 \cdot 10^{-7}$ (script `05_compare_source.py`), consistent with floating-point quadrature.

	$\beta = 0$		$\beta = 1$		$\beta = 2$		$\beta = 5$		$\beta = 10$	
ν	η_0	η_2	η_0	η_2	η_0	η_2	η_0	η_2	η_0	η_2
1/9	0.02016	−0.0349	0.03024	−0.1555	0.01472	−0.2905	$4.214 \cdot 10^{-4}$	−0.6745	$5.381 \cdot 10^{-7}$	−1.2886
1/4	0.07486	−0.0417	0.06297	−0.1600	0.02732	−0.2890	$9.934 \cdot 10^{-4}$	−0.6592	$2.224 \cdot 10^{-6}$	−1.2546
3/7	0.1649	−0.0483	0.1098	−0.1658	0.04751	−0.2909	0.002239	−0.6512	$8.628 \cdot 10^{-6}$	−1.2344
1/2	0.2041	−0.0506	0.1298	−0.1680	0.05668	−0.2920	0.002917	−0.6497	$1.343 \cdot 10^{-5}$	−1.2301
2/3	0.2997	−0.0552	0.1789	−0.1726	0.08016	−0.2948	0.004928	−0.6485	$3.226 \cdot 10^{-5}$	−1.2245
1	0.5	−0.0625	0.2847	−0.1803	0.1343	−0.3003	0.01078	−0.6502	$1.187 \cdot 10^{-4}$	−1.2232
3/2	0.8103	−0.0706	0.456	−0.1889	0.2283	−0.3075	0.02387	−0.6559	$4.378 \cdot 10^{-4}$	−1.2291
2	1.125	−0.0766	0.6366	−0.1952	0.3323	−0.3133	0.04132	−0.6617	0.001058	−1.2367
7/3	1.336	−0.0799	0.7604	−0.1985	0.4053	−0.3164	0.05487	−0.6652	0.001657	−1.2415
4	2.395	−0.0907	1.4	−0.2094	0.7951	−0.3271	0.1376	−0.6777	0.006706	−1.2594
9	5.578	−0.1035	3.389	−0.2219	2.044	−0.3398	0.4413	−0.6929	0.03318	−1.2806

Table 25 / Certified critical removal rates η_0 and direction coefficients η_2 (midpoints of the enclosures, rounded; the full enclosures for all 1111 points are in `certificates/grid_results.csv`). Every η_2 is certified negative.

7 The case $\nu = 0$, $\beta = 3$: proof of Theorem 8

For $\nu = 0$ and odd β the source finds infinitely many crossings, with H decaying super-exponentially. At the eleventh zero, $|H'| \approx 10^{-8}$, and a plain ball evaluation over a t-interval cannot resolve the sign of H, because its overestimation is of order 1 times the width. We therefore use, for $\beta = 3$,

$$H(t) = \frac{1}{2}\int_{\mathbb{R}} e^{-y^4/4 - ity}\,dy = \frac{1}{2}\int_{\mathbb{R}} e^{-(x-ic)^4/4 - it(x-ic)}\,dx, \qquad c = \tfrac{1}{2}\,t_c^{1/3},$$

(Cauchy's theorem; the shifted line passes through the saddle points $t^{1/3}e^{-i\pi/6}$ and $t^{1/3}e^{-5i\pi/6}$), and the same for H' with the factor $-iy$. On this line the integrand has the size of H itself, so the enclosures have small relative error. For $|x| \ge X \ge \sqrt{12}\,c$, $\operatorname{Re}(x - ic)^4 \ge x^4/2$, so the tails are at most $e^{-tc}(1 + c/X)\,2e^{-X^4/8}X^{-(3-m)}$ ($m = 0, 1$ for H, H'). Script `04_odd_beta3.py` certifies: 17 zeros in $(0, T_{\mathrm{scan}}]$, $T_{\mathrm{scan}} = 27.77$, by sign changes and a constant sign of H' around each zero; that H has no other zero there, by the centred form $H(J) \subset H(m) + H'(J)[-w/2, w/2]$ on the gaps; $\operatorname{Im} G(it_j) < 0$; that $\vartheta_1 > \cdots > \vartheta_{11}$ and $\vartheta_j < \vartheta_{11}$ for $j = 12, \dots, 17$; and $T_{\mathrm{scan}}^2\vartheta_{11} > 2$. For $\nu = 0$ an integration by parts gives $G(it) = \frac{1}{it}\,(1 + \int_0^\infty e^{-ity}\phi'(y)dy)$ with ϕ decreasing from $\phi(0) = 1$, so $|G(it)| \le 2/t$, and no imaginary root with $\vartheta \ge \vartheta_{11}$ has $t > T_{\mathrm{scan}}$ ($\vartheta t \ge \vartheta_{11}T_{\mathrm{scan}} > 2/T_{\mathrm{scan}}$). This gives the root picture of Theorem 8, (H3) at every t_j ($j \le 11$), and (H4) at t_1. The argument of Proposition 20, applied between consecutive ϑ_j, gives the alternation of $N(\vartheta)$ between 0 and 2, since the crossing directions $\operatorname{sgn} H'(t_j)$ alternate. The values are in Table 26; $\eta_2 < 0$ at every crossing.

8 Scope and remarks on the source

- *Model.* Everything concerns the limit model (1), equivalently (E_η), in which the source observes the oscillations. The derivation of (1) from the Becker–Döring system with injection and depletion is formal in the source, and a Hopf theorem for that infinite system is not attempted here.

j	$\eta_{(j)}$	$\kappa_{(j)}$	η_2	$\operatorname{Re}\hat{g}_1(1)$
1	$2.309572392 \cdot 10^{-3}$	0.5353315585	−0.43548182	76.112489
2	$5.283502051 \cdot 10^{-7}$	0.1293368539	−0.27019151	−728.17050
3	$9.258695924 \cdot 10^{-8}$	0.1188537500	−0.34140296	981.31501
4	$1.143390290 \cdot 10^{-8}$	0.08941878885	−0.33136298	−670.81795
5	$2.802774565 \cdot 10^{-9}$	0.07537569842	−0.33354598	323.96458
6	$8.699904338 \cdot 10^{-10}$	0.06507324666	−0.33321575	−126.62050
7	$3.288440931 \cdot 10^{-10}$	0.05763377403	−0.33330649	42.917776
8	$1.422225926 \cdot 10^{-10}$	0.05190101582	−0.33330715	−13.129710
9	$6.818335235 \cdot 10^{-11}$	0.04734503300	−0.33331673	3.7179564
10	$3.541935337 \cdot 10^{-11}$	0.04362387789	−0.3333211	−0.9911355
11	$1.962498289 \cdot 10^{-11}$	0.04052039391	−0.333324	0.2517176

Table 26 / The eleven certified crossings for $\nu = 0$, $\beta = 3$ (midpoints; all enclosures have radius below 10^{-6} relative, see `certificates/odd_beta3.csv`).

- *Parameters.* For $\beta = 0$ the result covers every $\nu > 0$. For $\beta > 0$ it covers the 1111 grid points of Theorem 7, which include all values of β in the source's Tables 2–3 and Figures 3–4 for its nine values of ν, and the eleven crossings of its Table 1. Between grid points the hypotheses are expected to hold by continuity, but this is not proved. A centred-form version of the certificate over β-boxes works for small ν, but for large ν the ball evaluation of the oscillatory integrals with a ball-valued β overestimates the β-dependence by factors of 10^2 to 10^4 (at $\nu = 9$, boxes of width $2 \cdot 10^{-4}$ were still needed), so a cover of the whole interval $[0, 10]$ would need higher-order Taylor forms. This was not carried out.
- *Stability.* Part (5) of Theorem 5 concerns the characteristic roots of the steady state. Nonlinear stability of the steady state for $\eta > \eta_0$ and orbital stability of the bifurcating orbits are not proved.
- *Normalisation.* See Remark 11: in the variables of (1) the critical rate is $\eta_0/(1-\alpha)^\beta$.
- *Table 1 of the source.* Its column κ_2 equals $\eta_2/4 = \eta_2/(\beta+1)$ in every row, which disagrees with its own formula (3.23). For example, at the first crossing the certified value is $\kappa_2 = -0.10842900 \pm 3 \cdot 10^{-9}$, whereas Table 1 gives −0.10887. A floating-point Fourier–Galerkin continuation of the branch (`06_galerkin_check.py`, not part of the proof) agrees with the certified values: $\kappa_2 \approx -0.108429$. Its η_2 values and those of the main data files agree with ours.

9 Computer verification

All scripts are in `scripts/` and use Python 3 with python-flint 0.9 (Arb). Each prints `PASS`/`FAIL` per claim and exits non-zero on failure. Scripts 01, 03 and 04 were run under WSL Fedora and again under Windows (Python 3.12, python-flint 0.9.0), with identical outcomes; the logs are in `certificates/`.

`bdhopf.py` Enclosures of G (Lemma 24), of H, H' on the saddle line for $\beta = 3$, and of the Hopf quantities (2)–(4).

`certify_lib.py` Checks (C1)–(C5) of Section 6.2.

`01_beta0_direction.py` Lemma 23 and the consistency check of Proposition 22 (one minute).

`02_search_grid.py` Floating-point search of the intervals T (exploration; writes `certificates/grid_tintervals.csv`).

`03_certify_grid.py` Theorem 7: (C1)–(C5) at the 1111 points (3 to 20 minutes on 8 cores, depending on the load); writes `certificates/grid_results.csv`.

`04_odd_beta3.py` Theorem 8 (about 2 minutes).

`05_compare_source.py` Comparison with the source's data files and the table of Table 25 (not part of the proof).

`06_galerkin_check.py` Floating-point continuation of the branch at small amplitude (not part of the proof).

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